

Project Overview

Project: Logistics Empire

Role: Systems Designer (2024 – present)

Released: 2025 – present

Format: Web-based Logistics Simulation

Summary: Logistics Empire is a logistics simulation built on real-world maps and street routing. Players build their own logistics business and transport more than 100 resource types between geographically accurate trade partners, ranging from supermarkets to wholesale distributors.

H3 Hex Map

Overview

Logistics Empire uses a hexagonal map for building placement, with each tile type derived from real-world geography and spatially indexed using H3.

To create this map, I researched and categorized geographic data from OpenStreetMap and designed a method for translating its geographic data into gameplay-relevant hex tile classifications.

Each building can only be placed on specific tile types, such as industrial, residential, or agricultural. The underlying geospatial data therefore determines where buildings can be placed.

My contribution: OSM research and categorization · Overpass queries · classification rules · weighting and edge-case design · collaboration with client/server developers · testing and iteration



Design Challenge

OpenStreetMap provides a vast public dataset covering many different types of geographic information. Beyond its scale, a key challenge is regional variation in mapping practices, meaning that a global classification system must account for inconsistent tagging and local conventions.

For the simulation, we needed to filter, categorize, and reduce this data into a smaller, unified dataset. This allowed the game to evaluate actions such as building placement efficiently and consistently, while supporting global use through a single ruleset.

Rendering the complete map was time-consuming, so we also needed a way to test and evaluate classification rules before generating the final map.

Design Goals

The resulting map should feel plausible to players and remain consistent with the street map displayed beneath it.

Players should be able to intuitively predict where different buildings can be placed—for example, farms in rural areas and factories in urban or industrial areas.

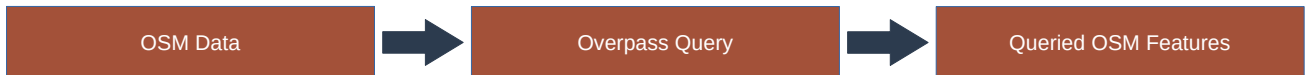
The classification needed to work globally, allowing players to use the same system anywhere in the world. It also needed to be reproducible, so rerendering the map after data updates would produce broadly consistent results.

The number of tile types should remain relatively small, allowing players to build a clear mental model of the map and understand their surroundings.

Data / Research

We chose H3 because it provides an efficient way to index and query large spatial datasets. Its hierarchical structure also allows the map to use a coarser H3 resolution while zoomed out, improving rendering performance while still providing a meaningful summary of the underlying data.

I used Overpass to query OpenStreetMap and identify features representative of different land types – for example farms, fields, and meadows for agricultural tiles. The resulting queries and categorized feature lists were then passed to our server and native client developers for data export and map generation.



Solution

OpenStreetMap primarily represents geographic information through nodes, ways, and relations. Ways are composed of nodes, while relations group nodes and ways. For classification, we resolved relevant ways and relations into spatial inputs that could be evaluated against the H3 cells.

For each hex tile, the algorithm counted relevant nodes and calculated a score for each category. This produced reasonable results, but some feature types naturally appeared far more often than others, even when a less numerous feature was more significant – for example, a large factory being outweighed by a row of individually mapped street trees.



Weights

To improve the classification, I worked with our native client developer to introduce weights for different feature types, allowing less numerous but more significant features to contribute more strongly to the result. We also evaluated the area of closed relations so that large features could receive appropriate weight even when they contained few nodes or extended beyond the tile.

Empty Tiles

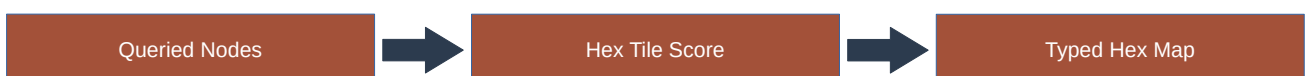
Some tiles contained no relevant features and therefore remained unclassified. In cases such as oceans or deserts, this was intentional, but other tiles clearly belonged to a category – for example, a tile in the middle of a large forest whose polygon boundary lay entirely outside the tile.

To address this, we also evaluated relations whose areas overlapped the tile, even when their defining nodes were located outside it.

Restricted Tiles

We restricted tiles representing locations that were excluded from player construction, such as cemeteries, religious sites, nature reserves, and military areas.

Other locations, including harbors, airports, and train stations, were reserved for future features.



Testing

To test the resulting tile distribution, we rendered selected map regions with each classification shown in a different color. I asked testers to review areas they knew well and identify classifications that did not match their expectations.

Their familiarity with these locations provided valuable feedback on how players would perceive the map. I used these observations to refine feature weights and improve the alignment between the generated classifications and recognizable real-world land use.

Example – Industrial Tiles

The following example shows how I developed the classification rules for industrial tiles.

Research

To identify suitable OSM features, I reviewed the OpenStreetMap documentation and inspected known industrial areas using OSM query tools. This allowed me to compare how industrial locations were tagged across different regions and identify combinations that reliably represented industrial land use.

The classification was primarily data-driven, but also needed to reflect how players would intuitively perceive an area when comparing the hex map with the underlying street map. This research resulted in the following Overpass query.

Query Extract

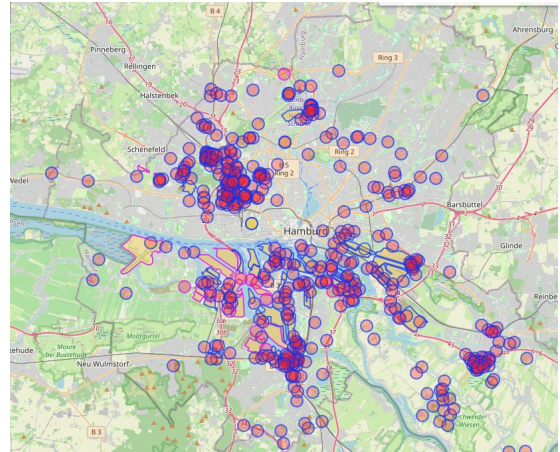
```
nwr(area.boundaryarea) [
    industrial~"factory|
        warehouse|
        brickyard|
        sawmill|
        distributor|
        brickworks|
        metal_processing|
        refinery|
        chemical|
        logistics|
        depot|
        cooling|
        storage|
        machine_shop| "
];
nwr(area.boundaryarea) [
    landuse~"industrial"
];
```

Query Explanation

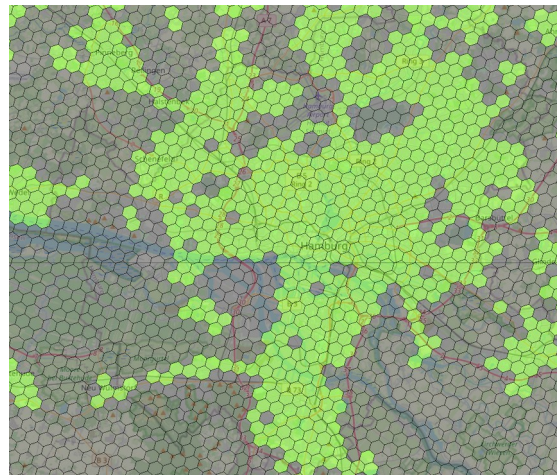
The query focuses on combinations of the OSM tags as **industrial** and **landuse**, together with values such as factory and industrial, to identify features that reliably indicate industrial land use.

Using combinations of tags rather than broad individual tags reduced false positives. For example, the industrial key can also contain features unrelated to industrial land use, such as communication infrastructure. Explicitly filtering accepted values also made the classification more robust against unrelated values being added to OSM later.

Note: The rendered hex map also contains other tile classifications, as buildings are generally not restricted to a single tile type.



Overpass Query Result – Hamburg Region



Generated H3 Tile Map Rendering

Trade-offs

The main trade-off was that each tile could contain only one type, as the existing rendering system supported only a single classification per tile.

This single-type approach inevitably created ambiguous results when category scores were close. A tile that was 49% industrial and 51% forest, for example, would be classified entirely as forest.

The underlying OpenStreetMap data is also incomplete and inconsistent in some regions, and certain types of geographic information are either rarely mapped or not represented at all.

For Logistics Empire, this simplified approximation was sufficient because building-placement rules could remain flexible, with some buildings allowed on multiple tile types.

Outcome

The hex map launched with Logistics Empire and remains in active use, with new classifications added as needed. The system currently covers the entire globe, with around 20 tile categories supporting placement rules for more than 180 buildings.

To address some of the original limitations and support future content, the system is being expanded to support multiple types per tile, manual classification edits, and regional patching to reduce rerendering time.